

CRYOGEN FREE MEASUREMENT SYSTEM



Made by - **Guided by -**
Pranav Choubey Dr. Sanjay Singh

School of Material Science and
Technology,
IIT (BHU) Varanasi.

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Introduction to Cryogen-Free Measurement Systems (CFMS)

The Cryogen-Free Measurement System (CFMS) developed by Cryogenic Limited is an advanced, helium-less platform used for very high magnetic field and low temperature experiments. It is specifically engineered to reduce the requirement of liquid helium which is major challenge in low temperature research.

Purpose of CFMS:

- To study transport properties (I-V characteristics, resistivity, magnetoresistance, Hall effect) of materials under extreme conditions of temperature and magnetic field.
- Designed for ease of operation, reliability, and accurate measurements over long durations.
- The mCFMS also known as Mini High Field Measurement CFMS (3–7 T range) offers a self-contained system with a pulse-tube cryocooler, superconducting magnet, and variable temperature insert (VTI).

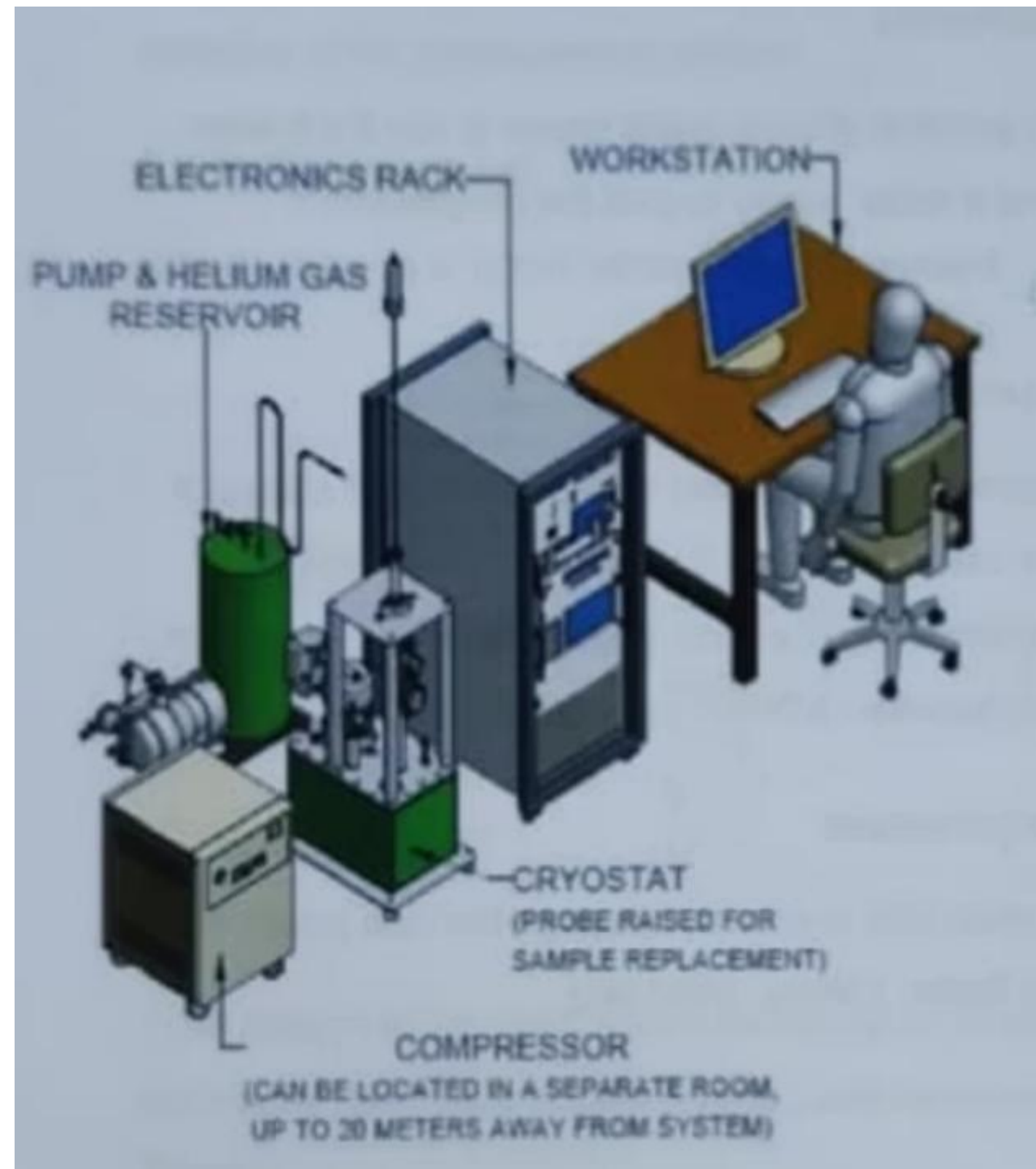
Components of CFMS:

The CFMS is composed of five major sub-systems:

1. Cryo-cooler system
2. Cryostat and superconducting magnet
3. Variable Temperature Insert (VTI)
4. Electronics Rack
5. Measurement Software (LabVIEW-based)

Each of these subsystems works together to achieve the goal of delivering a flexible cryogenic experimental platform.

Sufficient LAB space and Air-Conditioned is required of the CFMS setup.



Cryo-Cooler System

Overview:

A mechanical refrigeration device that cools system components using non-consumed helium gas.

Types of Cryocoolers:

1) Gifford-McMahon (GM):

- 1) High thermodynamic efficiency.
- 2) Uses pistons and regenerators.
- 3) Operates in various orientations.

2) Pulse Tube (PT):

- 1) No cold moving parts.
- 2) Quieter, more reliable.
- 3) Longer service intervals, lower operating costs.

Functionality:

- Stage 1 (40-60 K): Cools the radiation shield.
- Stage 2 (4 K): Directly cools the magnet and Variable Temperature Insert (VTI).

Cooling Performance:

- 50 W(J/s) at 60 K
- Up to 1.5 W(J/s) at 4 K
- (i.e. 1.5 J of heat is absorbed per second by cryocooler at a temperature of 4 K)

Construction:

- Cold head on cryostat top plate.
- Connected to helium compressor via high-pressure hoses (up to 20 m).



**F-50 Sumitomo Cryogenics
Pulse Tube Cryo-Cooler**

Working of Cryo-Cooler

- Helium returns from cryocooler at $\sim 1\text{--}2$ MPa.
- Helium enters sealed scroll compressor; orbiting scroll compresses helium by shrinking gas pockets.
- Oil lubricates, seals gaps, and absorbs heat during compression.
- Compressed helium ($\sim 18\text{--}25$ bar) exits mixed with oil.
- Bulk oil separator removes large oil droplets by gravity.
- Gas-oil mixture is cooled in water-cooled heat exchanger.
- Cooled oil is reinjected into scroll; excess oil returns to sump.
- Fine mist separator removes remaining oil droplets.
- Adsorber removes final oil vapor and moisture.
- Clean, dry, high-pressure helium goes to cryocooler.
- Relief valves, equalization valve, and sensors protect and monitor system.



Cooling Procedure:

- **The Chiller** cools water using compressors inside it to 17–20 °C.
- This water cools the **helium compressor** of Cryo Cooler.
- **Cryo-Cooler** compresses helium gas → generates high-pressure helium.
- High-pressure helium flows to the **cold head**.
- Inside cold head: **Regenerator** pre-cools helium gas.
- Then gas enters the **Pulse Tube**, where it **expands and cools**.
- Cold helium (~4 K) absorbs heat from **magnet/sample**.
- Warmed low-pressure gas returns to compressor.
- **Cycle repeats** for continuous cooling.
- **Chiller** removes heat from compressor, and fan ejects it to air.



Water Chiller

Cryostat and Magnet

Cryostat:

- A vacuum-insulated enclosure.
- Provides mechanical support and thermal isolation.
- Houses the superconducting solenoid and the VTI.

Magnet:

- A vertically oriented filamentary superconducting solenoid.
- Made from:
 - NbTi wires for lower field strengths.
 - NbTi + Nb₃Sn wires for higher fields.
- Copper stabilized for safety and performance.
- Connects electrically via High-Temperature Superconducting (HTS) current leads to minimize thermal conduction from the outside.

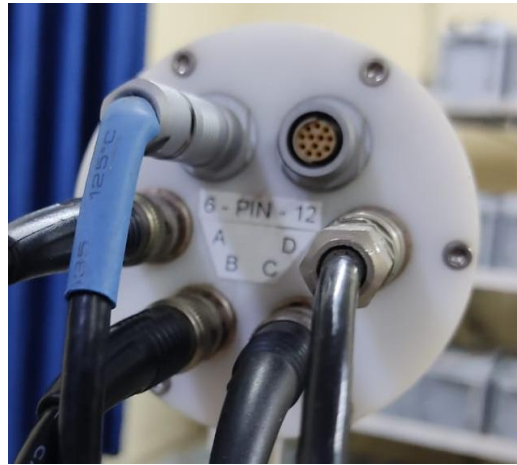


**Cryogenics
Limited mCFMS
Cryostat and
Magnet**

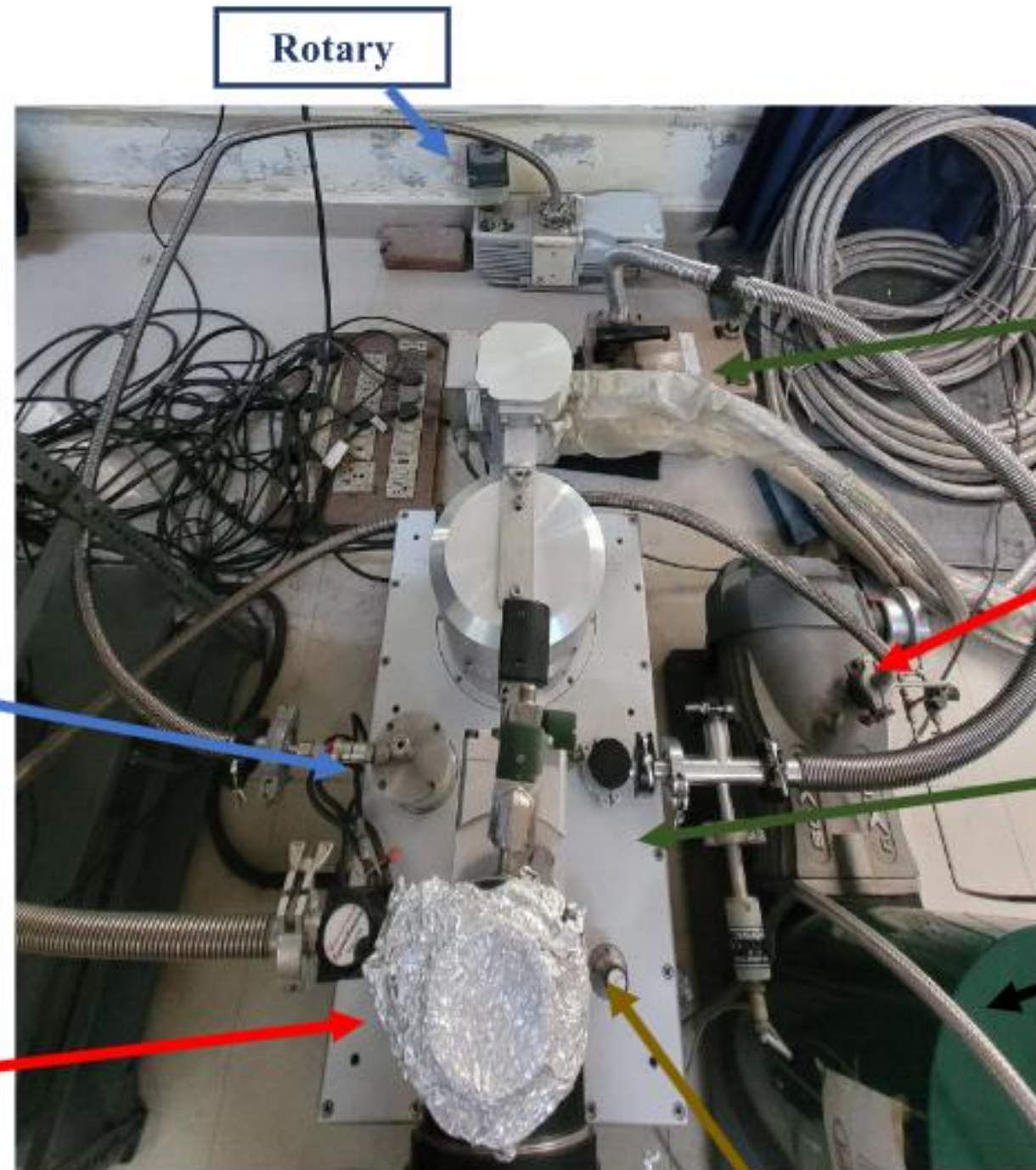


**Helium
Gas
Reservoir**

Visual Representation of Machines Used in CFMS



Probe



Rotary

High vacuum
Rotary

Scroll Pump

Chamber

He-dump

He-inlet

VTI

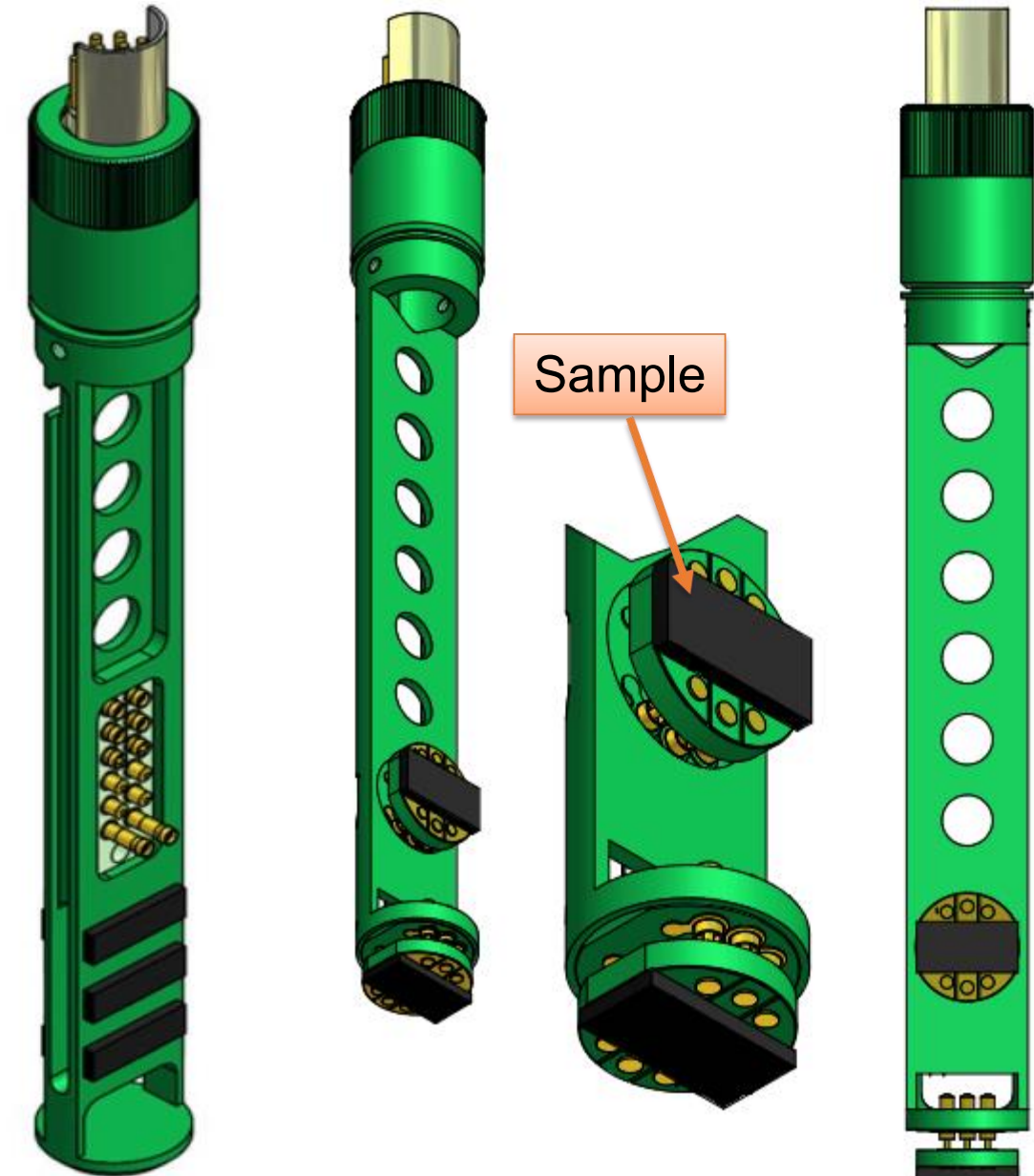
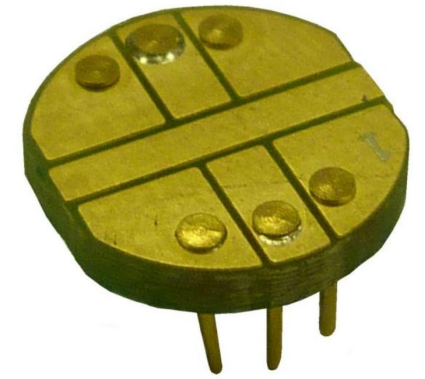
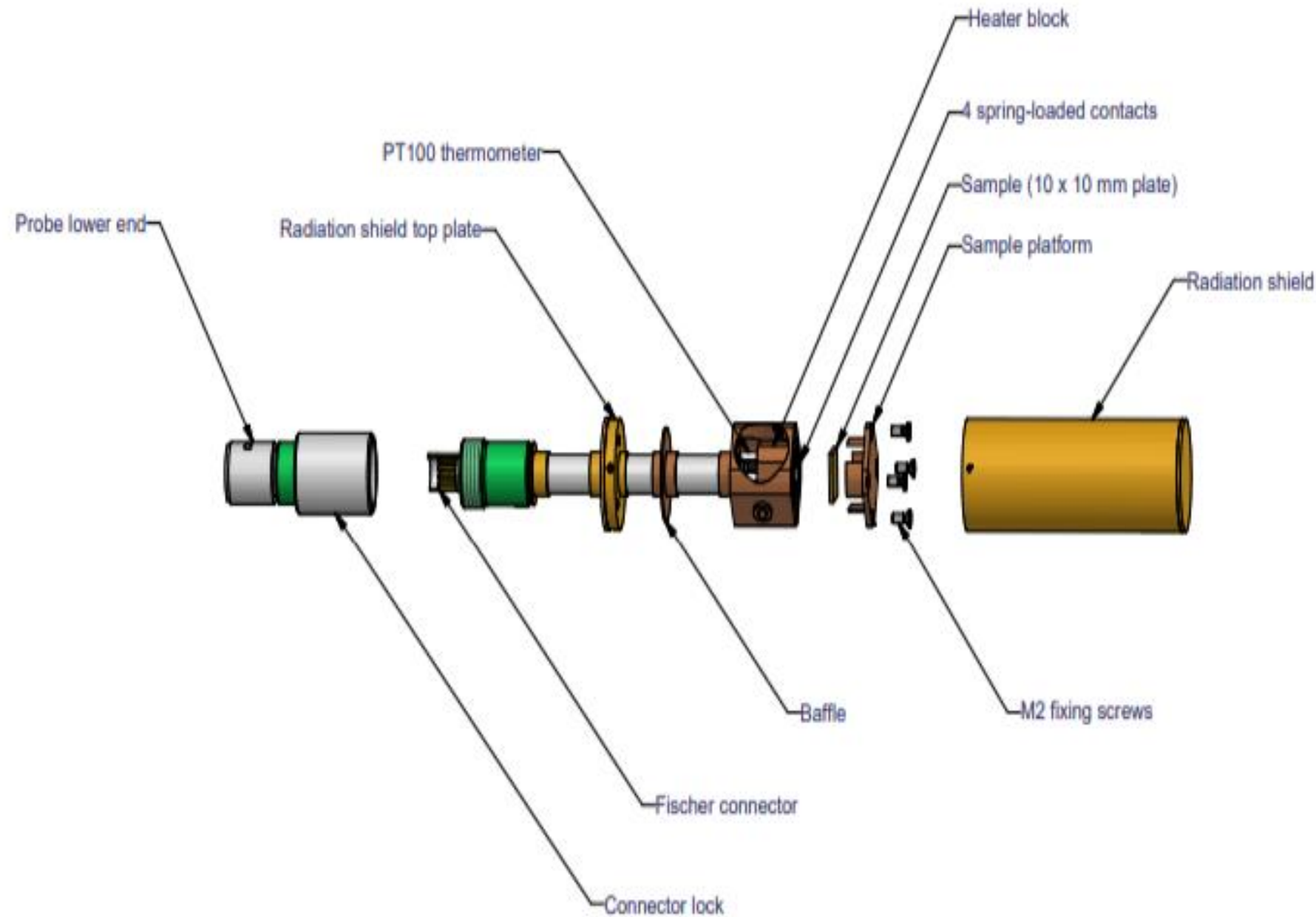
Needle valve



Pressure Guage



Sample holder diagram



Variable Temperature Insert (VTI)

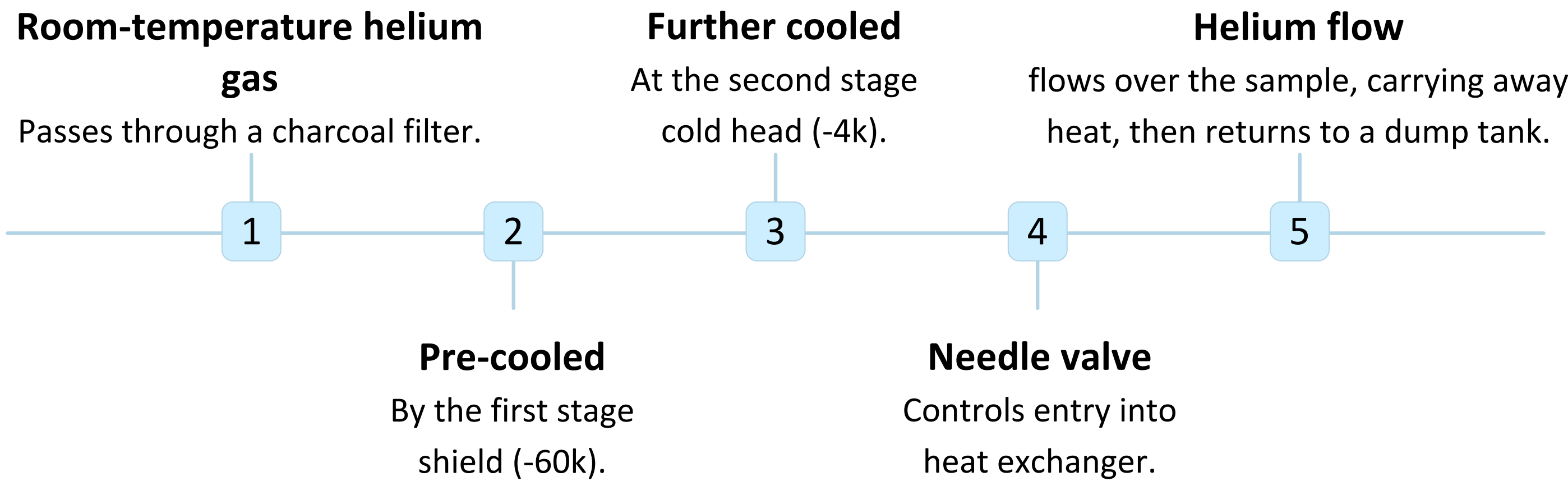
Function:

- 1

Controls sample space temperature over a wide range.
- 2

Enables precise cooling via regulated helium gas flow using a needle valve.

Operational Sequence:



Thermometry:

1 Cernox Sensor A

At the heat exchanger (for general system temperature control).

2 Cernox Sensor B

At or near the sample platform (for direct sample temperature).

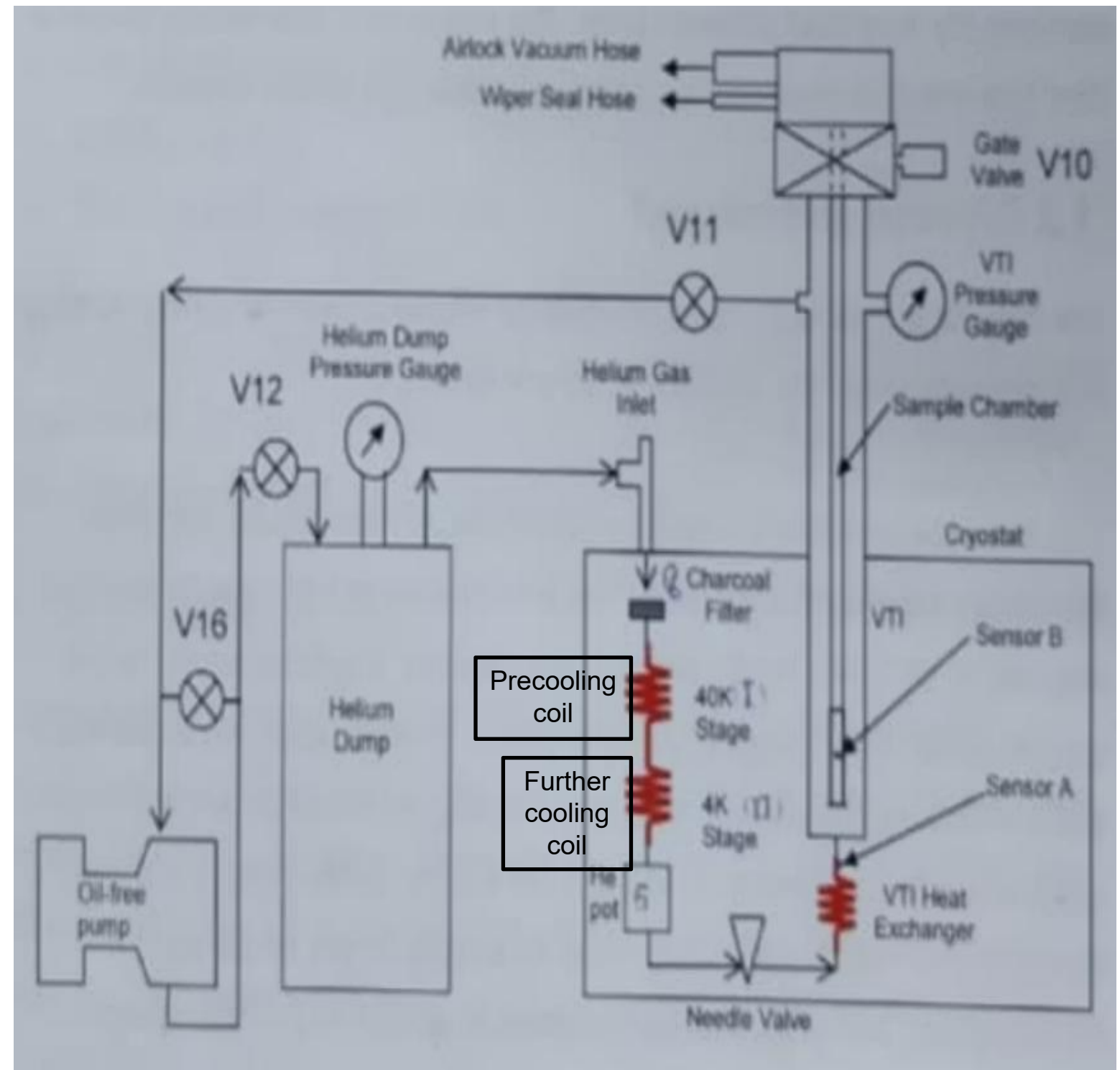
Additional Features:

1 Dual heating

- One at heat chamber
- One near the sample.

2 Integrated

with feedback-controlled heaters for precise temperature ramps and stability.



Schematic Diagram of VTI Cooling Circuit

Vacuum Pumps

There are two types of vacuum pumps used for transport of helium in cryogen free measurement system:

A) Oil-Free Scroll Pump

- Two spiral scrolls: one fixed, one orbiting.
- Gas enters at the edge, gets trapped in pockets.
- Pockets move inward as volume decreases, compressing gas.
- Compressed gas exits through center exhaust.
- No oil needed → clean, low-maintenance, ideal for CFMS.

B) Oiled Rotary Vane Pump

- Rotor with spring-loaded vanes rotates inside chamber.
- Gas enters, trapped between vanes, compressed as rotor turns.
- Compressed gas exits through exhaust port.
- Requires oil for sealing, lubrication, and cooling.
- Higher maintenance, risk of oil contamination.

Both the pumps have a vacuum range of nearly 10^{-3} mbar.



Oil-Free Scroll Pump



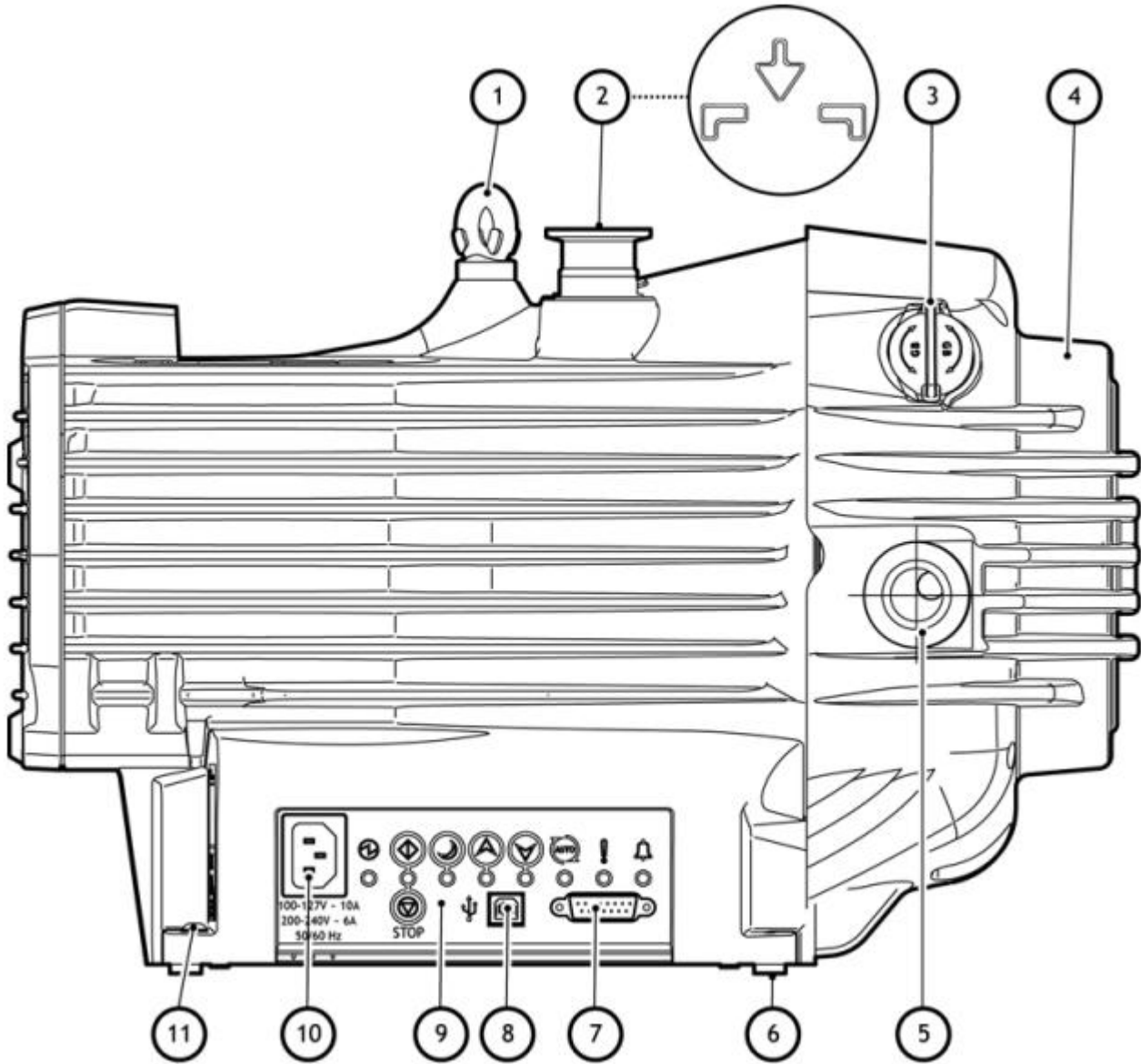
Oiled Rotary Vane Pump

Edwards nXDS 6iR Pump

Code	Meaning
nXDS	Edwards scroll pump series name (Next generation X Dry Scroll)
6	Pumping speed: 6 m ³ /h (cubic meters per hour)
i	"Intelligent" — has advanced control features, digital interface, better monitoring
R	"Corrosive Resistant" version — designed with special internal coatings/seals to handle light corrosive gases

Bearing: There exists a set of bearing inside the scroll pump which helps in:

- Maintaining orbital shaft's precise orbiting motion.
- Keep the correct alignment between orbiting and fixed scrolls.
- Reduce friction and mechanical wear.



1. Lifting eye

2. NW25 inlet port

3. Gas ballast control

4. Cooling fan

5. NW25 exhaust port

6. Rubber feet
7. 15-way D-type connector

8. USB port (service mode only)

9. User interface panel

10. Mains power connector

11. Secondary earth bond point

Electronics Rack

It consists of all the electronic hardware and machines which comprises the cryogen free management system. The components of electronics rack are:

- Temperature Controller
- Temperature Monitor
- Magnet Power Supply
- Nanovoltmeter
- DC and AC Current Source
- Two Computers
- CPU for Computers



Electronics Rack

A) Temperature Controller and Monitor

1) Lake Shore Model 350 Temperature Controller

- **Function:** Controls and monitors temperature.
- **Inputs:** Up to 4 cryogenic sensors (Cernox, diode, etc.).
- **Output:** Heater control (up to 75 W).
- **Working:** Reads temperature → compares to setpoint → adjusts heater power using PID control.
- **Use:** Actively maintains sample temperature in cryogenic systems.

2) Lake Shore Model 218 Temperature Monitor

- **Function:** Monitors (only reads) temperature.
- **Inputs:** 8 sensor inputs (diode, thermocouple, RTD, etc.).
- **Output:** No heater control.
- **Working:** Continuously reads and displays temperatures.
- **Use:** Logs temperatures at multiple points (e.g. cryocooler, chamber walls).



Lakeshore Temperature Controller and Monitor

B) Cryogenic Magnet Power Supply

- Supplies precise DC current to superconducting magnet → creates magnetic field.
- Field strength $\propto$ current ($B = \mu_0 n I$).
- Uses slow ramping to prevent quenching.
- Supports both positive and negative currents (field reversal).
- Can operate in persistent mode (constant field without continuous power).
- Includes quench protection, overcurrent/overvoltage safety, and interlocks.
- Controlled manually or via computer (Using LabVIEW Magnet Power Supply VI).
- Works together with temperature control, vacuum, and measurement system.
- Typical specs: ~50–100 A, 5–10 V, ppm-level stability.



**Cryogenic SMS60C
Magnet Power Supply**

C) Nanovoltmeter and Current source

1) Keithley 2182A Nanovoltmeter

- Measures ultra-small voltages (nanovolt level).
- High input impedance, low noise.
- Filters external noise, stable readings.
- Used for Hall voltage, resistance, MR measurements.
- Syncs with 6221 for delta mode (noise cancellation).

2) Keithley 6221 DC/AC Current Source

- Supplies precise DC/AC current (pA to 100 mA).
- Very low noise, stable output.
- Generates AC waveforms for dynamic measurements.
- Paired with 2182A for precision measurements (delta mode).

Combined Together:

- 6221 applies current → 2182A measures voltage.
- Delta mode cancels thermal noise.
- Ideal for CFMS low-temp transport studies.



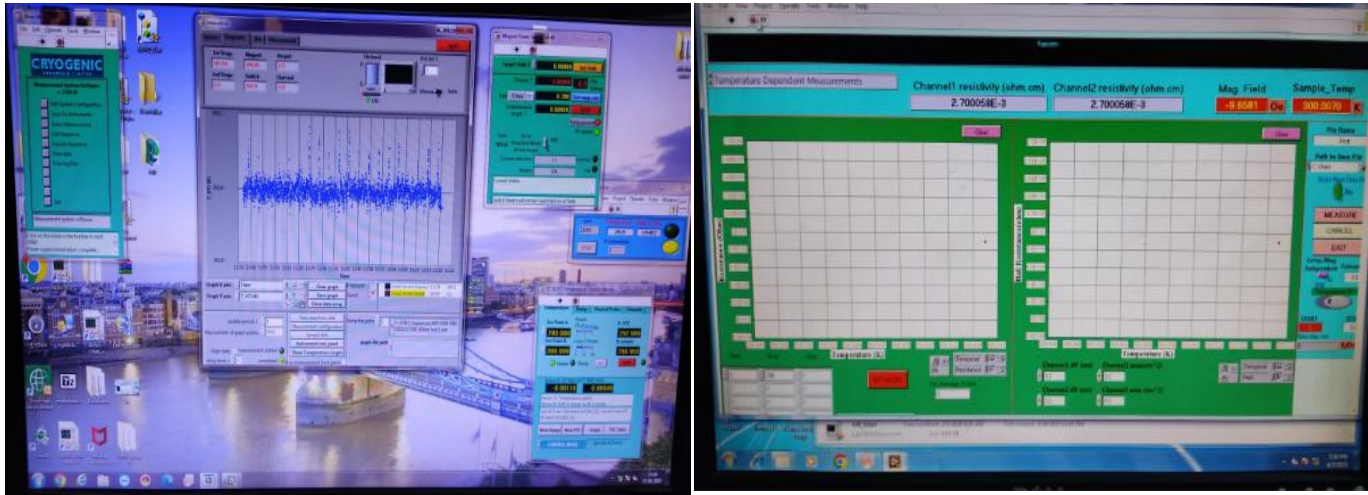
**Keithley 2182A
Nanovoltmeter**



**Keithley 6221 DC/AC
Current Source**

Software and System Control (LabVIEW)

The CFMS is operated via an integrated software suite built on LabVIEW, which provides full GUI-based control through modular Virtual Instruments (VIs).



System Computers



System Computers' CPU

Key Software Features:

- Built-in Sequence Editor for automated multi-step measurements.
- Real-time control over:
 - Temperature
 - Magnetic Field
 - Resistivity
 - Hall Voltage and Hall Resistivity
 - Valve States
- Real-time and historical data logging.
- Graphical representation of system state.

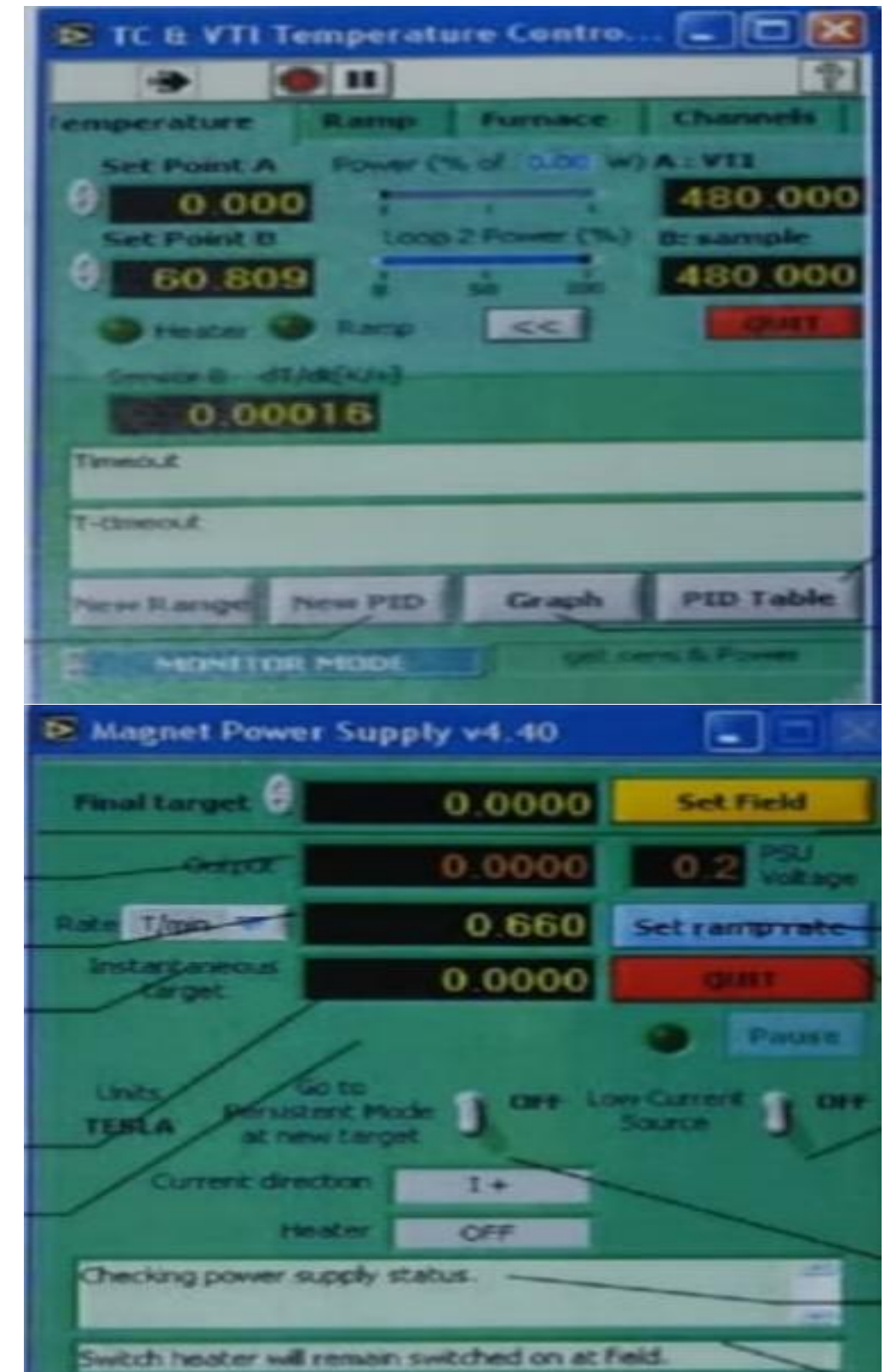
Virtual Instrument (VI) Modules

1. Temperature Controller VI

- Sets and maintains target temperatures for Sensor A and B.
- **Displays:**
 - Current temperature, Heater power and PID control status.
- **Interfaces:**
 - Temperature Control and Ramp Settings.

2. Magnet Power Supply VI

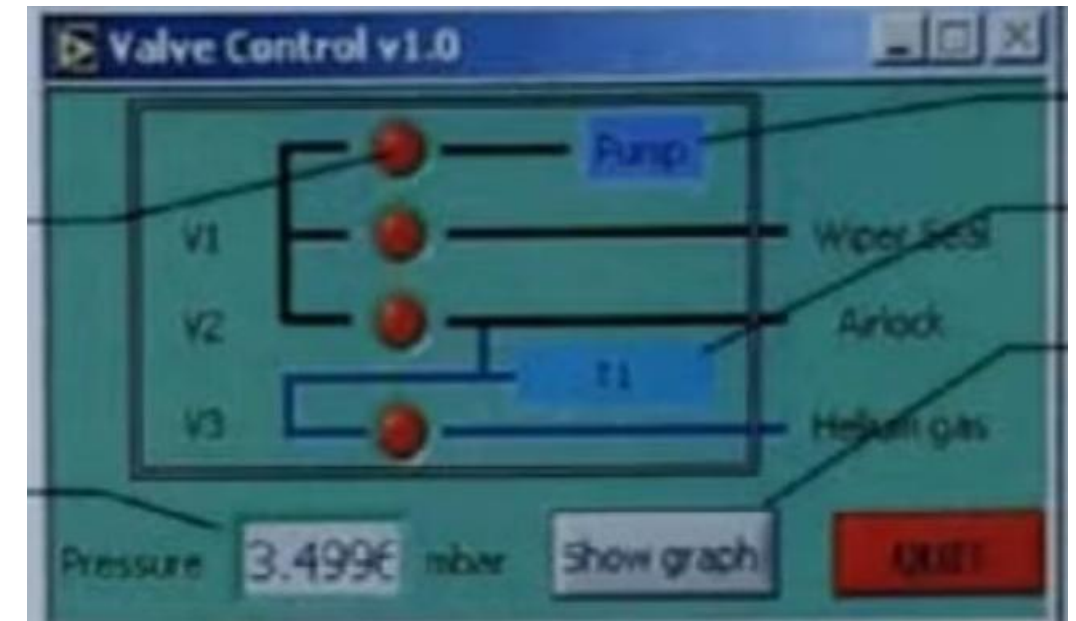
- **Displays and controls:**
 - Present field, Target field and Ramp rate.
- **Switches between:**
 - Non-persistent mode (heater ON, power supplied)
 - Persistent mode (heater OFF, field locked)



3. Valve Control VI

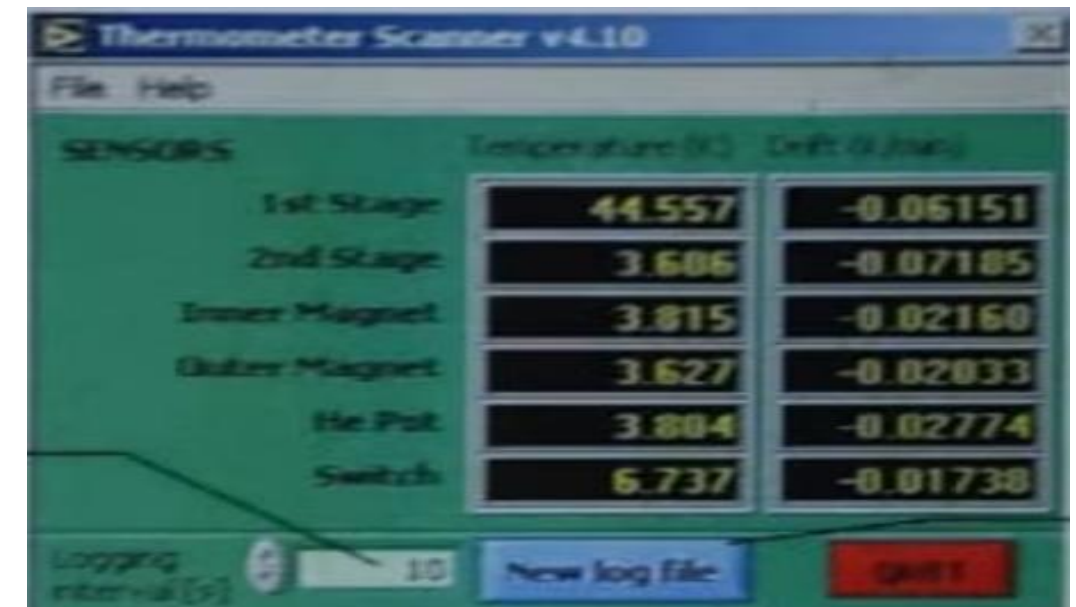
- Allows manual and automated control of cryogenic gas valves.
- Color-coded valve indicators:
 - Green = Open
 - Red = Closed
- Interlock logic prevents invalid configurations (e.g., opening inlet and exhaust simultaneously).

Note: We do the valve control manually in most experiments.



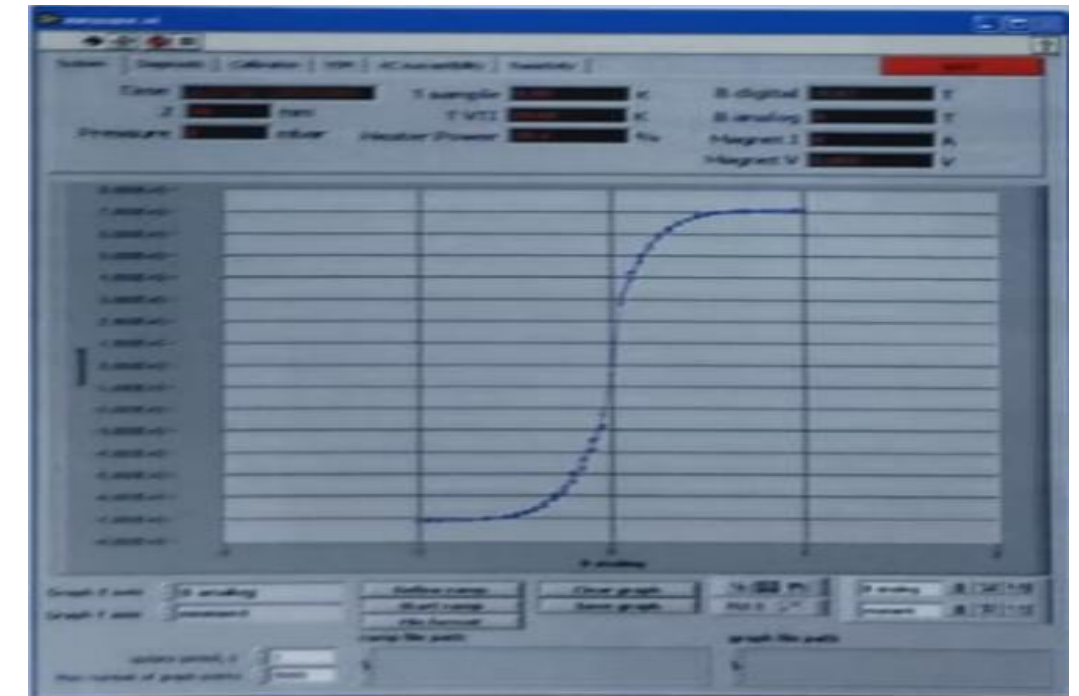
4. Thermometer Scanner VI

- Continuously logs temperature at six critical system points.
- Data logging functionality for cooldown diagnostics.
- Controlled via Lakeshore selector channel.



5. Measurement VI

- Controls 20+ internal variables.
- Ramp or step a variable (e.g., temperature, field).
- View previous data.
- Plot graphs and save experimental runs.

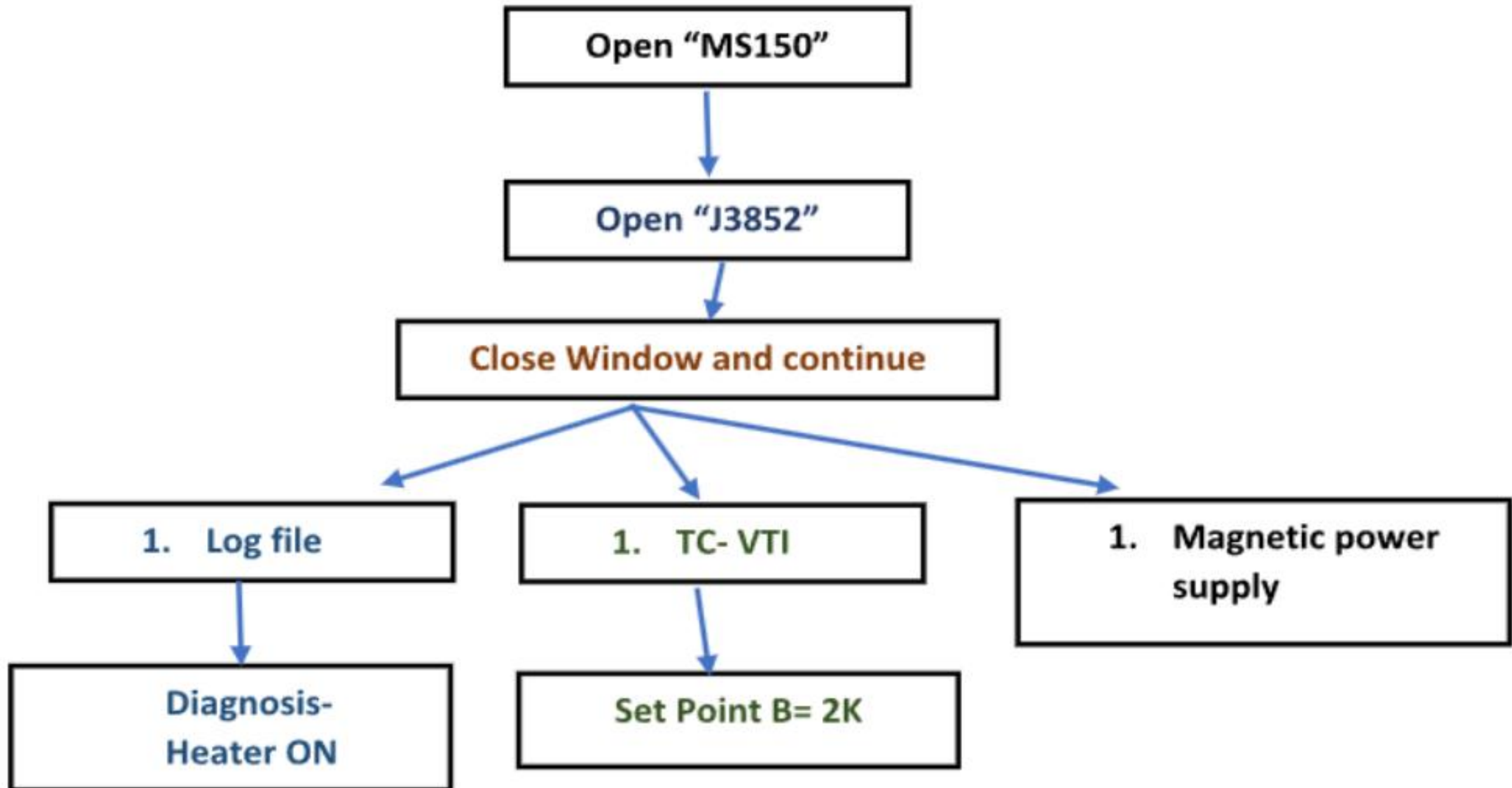


6. Sequence Editor VI

- Design and execute multi-frame, nested experiments.
- Include conditional logic and parallel tasks.
- Sequences stored in .seq format and reloaded on demand.



System Software Startup Procedure



Operating the system

(A) Vacuum-Purging

- Check temperature of chamber=Room temp ($\sim 300\text{K}$).
- Connect VTI- Scroll pump & He-inlet Rotary.
- Wait for 24-36 hours for temperature to remain stable.
- For vacuum of chamber:
 - (a) Close chamber valve (very important, so that oil from pump does not enter chamber.)
 - (b) Start high vacuum rotary.
 - (c) Check vacuum through pressure gauge (roughly ~ 0.001 mbar).
- Fill He in He-dump (till 0-0.5 psi).
- Close the He-dump valve.
- Connect He-dump to He-inlet and Rotary to the Scroll pump output. Open the Needle valve and then immediately open the Scroll pump valve.
- Wait for 24 hours.

(B) Cooling

- Switch "On" the Chiller.
- Fill He in He-dump till half (0-0.5psi).
- Then, open He-dump valve.
- Wait for 1-1.5 hours if system has been started after a long time, otherwise wait for 10-15 mins.
- Close the He-dump valve.
- Fill He in He-dump again (till 0-0.5 psi) and close the needle valve.
- Connect He-dump to He-inlet.
- Open Needle valve.
- Check the VTI Gauge: It should be between 10-15 mbar.
- Immediately close the scroll pump valve and open He-dump valve.
- Switch 'ON' the compressor.
- Check water level and the temperature of chiller regularly, the compressor starts working around 25°C.

(C) Cooldown

- Use thermometer scanner VI to confirm initial temperatures.
- Flush sample chamber and VTI.
- Load cooldown sequence in sequence editor.
- Begin sequence execution to automate ramp down.
- Monitor pressure and temperatures; helium dump should drop to nearly 0.2 bar.

(D) Measurement Execution

- Define sequence parameters: variable, ramp range, duration.
- Load and execute sequence.
- Live graph and data logging occurs.
- Parameters may include: temperature, field, valve states, etc.

Warming up the system

Procedure

- Open needle valve, switch off helium pump.
- Turn off compressor and wait 10 min before turning off cooling water.
- Allow system to warm **naturally (36–48 hours)**.
- Never disconnect helium dump or compressor while system is cold.
- Monitor using thermometer VI.

SAFETY GUIDELINES

Magnetic Field

Danger

Always maintain >2 m clearance from the magnet's center.

Magnetic Field Zone

Magnetic field danger zone exceeds 5 gauss. Remove all ferromagnetic tools, credit cards, medical implants

Overpressure

Protection

3 pressure relief valves: VTI, helium inlet, sample chamber.

Quench Protection

System stores high inductive energy. Never interrupt HTS leads unless magnet is fully discharged.

Maintenance Schedule

Weekly:

- Critically inspect thermometer readings.
- Grease mechanical machine parts with vacuum grease.

Annually:

- Call a certified engineer to inspect/repair cryo-cooler.
- Contact Cryogenic Ltd for planned service.

Troubleshooting Techniques

1

Issue: No VTI Pressure Change

Likely block due to helium contamination. Evacuate and re-flush the full VTI circuit.

2

Issue: Cryostat warm after cooldown

Check valve position. Inspect for ice formation. Redo helium circuit.

3

Issue: Magnet not stabilizing

Persistent switch heater might be defective. Ensure full field ramp before switching to persistent mode.

Applications of CFMS

- **Magnetic and Electrical Testing** – Study how materials react to magnetic fields and how electricity flows through them.
- **Superconductivity Research** – Analyze materials that conduct electricity without resistance at low temperatures.
- **Quantum and Phase Behavior** – Explore unusual effects and phase changes that occur in extreme cold or high magnetic fields.
- **Explaining important physics phenomena** –
 - I-V Characteristics
 - Resistivity
 - Hall effect
 - Magnetoresistance
 - Heusler alloys studies.

Keywords

A) Cernox Sensor:

- Type: *Resistance temperature sensor (RTD)*.
- Material: Zirconium oxynitride film on sapphire.
- Range: approximately 0.1 K to 420 K (excellent for cryogenics).
- High accuracy at low temperatures.
- Low magnetic field sensitivity.
- Small, easy to mount and fast response.
- Used in: Cryostats, CFMS, superconductivity, dilution refrigerators.

B) Delta Mode:

- Delta Mode is used for precise low-voltage measurements in *nanovoltmeter systems*.
- The current source reverses current direction; voltages are measured for both polarities.
- *Thermal EMFs* (caused by *Seebeck effect* due to temperature differences at junctions) stay constant; true signal changes sign.
- The nanovoltmeter calculates $(V_+ - V_-)/2$ to cancel thermal EMFs & low-frequency noise.
- This provides highly accurate, stable readings, ideal for cryogenic and transport measurements

System Specifications

1) GENERAL

Parameter	Value
Mains power requirements	240V, 50Hz, 5KW (maximum)

2) CRYOSTAT AND SUPPORT TABLE

Parameter	Value
Approximate Nominal weight (Kg)	150
Approximate Overall dimensions (m)	Width 0.45 × length 0.65 × height with probe raised 2.52

3) MAGNET

Parameter	Value
Magnet type	NbTi solenoid & persistent mode switch
Central field homogeneity %	0.2 over 20mm (± 10 mm)
Decay rate in persistent mode (ppm/hr)	10
Field direction	Vertical
Magnet inductance (Henries)	2.7
Maximum field (Tesla)	5 (G/A = 0.1065)
Nominal operating current (amps)	<60
Radial stray field 5 gauss contour (m)	0.7 (from centre of magnet)
Vertical stray field 5 gauss contour (m)	0.9 (from centre of magnet)
Initial magnet energisation rate (Tesla/min)	1
Maximum subsequent energisation rate (Tesla/min)	2
Magnet power supply	Cryogenic SMS60C (120A, ± 5 V)

4) CRYO-COOLER

Parameter	Value
Type	Sumitomo 0.5W Pulse Tube
Compressor mains requirements	380, 400, 415 V, 50 Hz, 3 phase, 13A operating current
Compressor cooling water minimum flow rate (l/min)	7
Static compressor gas pressure (MPa)	1.7 – 1.72
Working compressor gas pressure (MPa)	High: 2.1 – 2.5Low: 0.5 – 0.8
Typical cooldown time (hrs)	9

5) VTI

Parameter	Value
Temperature range (K)	1.6 – 325
Temperature stability (K)	± 0.05
Internal diameter (mm)	25
Normal operating pressure (mbar)	8
Dump vessel pressure with system at room temperature	Atmosphere plus 0.25 bar (approx)
Dump vessel pressure with VTI at base temperature	Atmosphere minus 0.75 bar (approx)

6) SYSTEM THERMOMETERS

Lakeshore Channel	Type and Label	Location and Typical Temperature
A	Sensor A: cernox	VTI heat exchanger
B	Sensor B: cernox	Within 10mm of sample position
C1	1st stage: ceramic composite sensor	Cryo-cooler 1st stage (25 – 40K)
C2	2nd stage: ceramic composite sensor	Cryo-cooler 2nd stage (3.0 – 4.5K)
C3	Magnet: ceramic composite sensor	Magnet winding (3.0 – 4.5K)
C4	Switch: ceramic composite sensor	Persistent mode switch (3.0 – 10.0K)
D1	He pot: ceramic composite sensor	Liquid helium pot (3.0 – 4.2K)
D4	Charcoal Trap: ceramic composite sensor	Charcoal filter (25 – 45K)

7) SAMPLE CHAMBER

Parameter	Value
Type	Isolated, filled with helium exchange gas
Internal diameter (mm)	25 mm (vacuum sample space)
Maximum sample size	4mm x 8mm